

Christina Shin

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INTERESTS	Computer Networking, Mobile Computing, AR/VR Systems, Volumetric Video Streaming, Autonomous & Connected Vehicles, 3D Data Processing, 3D Mapping	
EDUCATION	University of Southern California , Los Angeles, California	
	▪ Ph.D. in Computer Science	Jan 2026 (Expected)
	Ewha Womans University , Seoul, Republic of Korea	
	▪ M.S. in Computer Science and Engineering <i>Thesis: Network Diagnosis and Reconstruction in Vehicular Ad-Hoc Networks</i>	Feb 2019
EXPERIENCE	▪ B.S. in Computer Science and Engineering	Feb 2017
	General Motors , Warren, Michigan	
	Research Intern & Collaborator (Mentors: Fan Bai and Chuan Li)	May 2021 – Present
	▪ Led the design and implementation of a volumetric video delivery system for vehicles, enabling low-latency and immersive AR passenger experiences. ▪ Developed a 3D traffic reconstruction system using multi-vehicle point cloud registration, generating volumetric videos of dynamic traffic scenes to improve scalability and scene understanding; contributed to a U.S. patent application (US20230140324A1, now abandoned).	
	Networked Systems Lab , University of Southern California	
	Research Assistant (Advisor: Prof. Ramesh Govindan)	Aug 2019 – Present
	▪ Invented an infrastructure-assisted autonomous driving system that leverages roadside LiDARs and edge compute to extend vehicle perception beyond occlusions. ▪ Devised a 3D building reconstruction system with a LiDAR-equipped UAV, optimizing path planning for data capture and generating near real-time SLAM-based 3D models of buildings. ▪ Contributed to a live volumetric video streaming system, focusing on the 3D video rendering pipeline to ensure real-time visualization of captured scenes.	
	Intelligent Networked Systems Lab , Ewha Womans University	
	Research Assistant (Advisor: Prof. HyungJune Lee)	Jan 2017 – May 2019
	▪ Designed a traffic density estimation algorithm using opportunistic V2V packet probing under strict time deadlines. ▪ Proposed a multi-UAV relay algorithm for network recovery, improving resilience of ad-hoc networks in failure scenarios.	
PUBLICATION	CONFERENCE	
	▪ RECAP: 3D Traffic Reconstruction Christina Suyong Shin , Weiwu Pang, Chuan Li, Fan Bai, Fawad Ahmad, Jeongyeup Paek, and Ramesh Govindan <i>Proceedings of the ACM on Mobile Computing and Networking (MobiCom)</i> , 2024.	

- Cooperative Infrastructure Perception
Christina Suyong Shin*, Fawad Ahmad*, Weiwu Pang*, Branden Leong, Pradipta Ghosh, and Ramesh Govindan (* equal contributions)
IEEE/ACM Conference on Internet-of-Things Design and Implementation (IoTDI), 2024.
- AeroTraj: Trajectory Planning for Fast and Accurate 3D Reconstruction Using a Drone-based LiDAR
Fawad Ahmad, **Christina Suyong Shin**, Rajrup Ghosh, John D’Ambrosio, Eugene Chai, Karthikeyan Sundaresan, and Ramesh Govindan
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (Ubicomp/IMWUT), 2023.
- Progressive ad-hoc route reconstruction using distributed UAV relays after a large-scale failure
Christina Suyong Shin, So-Yeon Park, JinYi Yoon, and HyungJune Lee
IEEE Wireless Communications and Networking Conference (WCNC), 2018.
- DroneNet+: Adaptive Route Recovery Using Path Stitching of UAVs in Ad-Hoc Networks
So-Yeon Park, Dahee Jeong, **Christina Suyong Shin**, and HyungJune Lee
IEEE Global Communications Conference (GLOBECOM), 2017.

JOURNAL

- Infrastructure-less Vehicle Traffic Density Estimation via Distributed Packet Probing in V2V Network
Christina Suyong Shin, JiHo Lee, and HyungJune Lee
IEEE Transactions on Vehicular Technology (TVT), vol. 69, no. 10, Oct 2020.
- DroneNetX: Network Reconstruction through Connectivity Probing and Relay Deployment by Multiple UAVs in Ad-Hoc Networks
So-Yeon Park, **Christina Suyong Shin**, Dahee Jeong, and HyungJune Lee
IEEE Transactions on Vehicular Technology (TVT), vol. 67, no. 11, Nov 2018.

UNDER SUBMISSION

- Vehicular AR Streaming. *Submitted to MobiCom 2025.*
- Vehicular AR Tracking. *Submitted to MobiCom 2025.*
- Live Volumetric Video Streaming. *Submitted to CoNext 2025.*

AWARDS & HONORS

- Annenberg Fellowship, University of Southern California 2019
Awarded to outstanding Ph.D. students joining in Fall 2019.
- Qualcomm Innovation Awards, Qualcomm x Ewha 2017
Recognized for proposing a lightweight network hole replacement algorithm through UAV-net.
- Silver Prize in Graduation Capstone Design, Ewha Womans University 2016
For an outstanding project developing *SimMusic*, a language that encodes music notes and plays them on *Lego Mindstorms NXT*.
- Dean’s List, Ewha Womans University 2013, 2015, 2016
For attaining a GPA above 3.75/4.3.
- NSF/SIGMOBILE Travel Grants: ACM MobiCom ’25, ACM HotMobile ’24.

TEACHING & MENTORING

TEACHING ASSISTANT

University of Southern California

- Internetworking (CSCI 353) Fall 2025
- Computer Networks (CSCI 551/651) Fall 2023

Ewha Womans University

- Computer Architecture (20493-02) Fall 2018
- Arduino Programming (11208-01) Spring 2018
- C Programming (38407-05) Fall 2017
- Programming Language Theory (20499-01, 20499-02) Spring 2017

MENTORING

- Alejandra Cortes (High school): 3D perception with point clouds Summer 2023
- Alexander Due (Undergrad): Object-oriented point cloud registration Summer 2023
- Zaki Cole (Undergrad): Automated data collection in CARLA simulator Fall 2022

**TECHNICAL
SKILLS**

Languages: C++, Python, MATLAB, C, C#, Java

Systems & Tools: Linux, Git, Docker, ROS, CARLA, AirSim, Unity, PyTorch, TensorFlow